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Unit I

1. Introduction

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1.1 Introduction to Python:

Python is a high-level, general-purpose, and interpreted programming language developed by **Guido van Rossum** in **1991**. It is known for its simple, readable syntax that makes it easy for beginners to learn and powerful enough for advanced applications. Python supports multiple programming paradigms such as **procedural, object-oriented, and functional programming**.

Python comes with a large **standard library** and has extensive external libraries that support fields like **data science, web development, machine learning, automation, artificial intelligence, and scientific computing**. It is a platform-independent language, meaning Python programs can run on different operating systems like Windows, macOS, and Linux without modification. Overall, Python's simplicity, flexibility, and rich ecosystem make it one of the most widely used programming languages in the world today.

1.2 Features of python:

Python is a powerful and flexible programming language widely used in various fields due to its rich set of features. Some important features of Python are:

1. **Simple and Easy to Learn:**
   Python has a clean and readable syntax similar to English, which makes it easy for beginners to understand and write programs.

2. **Interpreted Language:**
   Python code is executed line by line by the interpreter. This makes debugging easier and allows rapid development.

3. **Platform Independent:**
   Python programs can run on different operating systems like Windows, Linux, and macOS without any changes, making it highly portable.

4. **Extensive Standard Library:**
   Python provides a large number of built-in modules and functions for tasks like file handling, networking, database access, and mathematical operations.

5. **Supports Multiple Programming Paradigms:**
   Python supports **procedural, object-oriented, and functional programming**, allowing developers to choose the style suitable for their applications.

These features make Python one of the most popular languages for application development, automation, data science, and emerging technologies.

1.3 What Can I Do with Python?

Python is a highly versatile programming language used across many domains due to its simplicity and powerful libraries. Some major applications include:

1. **Web Development:**
   Python is widely used to build dynamic websites and web applications using frameworks like **Django**, **Flask**, and **FastAPI**.

2. **Data Science and Machine Learning:**
   Python is the most popular language for data analysis, visualization, and AI development with libraries such as **NumPy**, **Pandas**, **Matplotlib**, **TensorFlow**, and **Scikit-Learn**.

3. **Automation and Scripting:**
   Python can automate repetitive tasks such as file handling, data entry, web scraping, and system administration, making it useful for productivity and DevOps.

4. **Software and Application Development:**
   Python is used to create desktop applications, GUI tools, and even games using libraries like **Tkinter**, **PyQt**, and **Pygame**.

5. **Networking, Cybersecurity, and IoT:**
   Python supports building network applications, writing security tools, and programming IoT devices using libraries like **Socket**, **Scapy**, and **MicroPython**.

Overall, Python's wide range of libraries and ease of use make it suitable for almost every modern technological field.

1.4 Python Interpreter:

A **Python Interpreter** is a program that reads and executes Python code line by line. It converts the human-readable Python program into machine-understandable instructions. Unlike compiled languages, Python does not require a separate compilation step; instead, the interpreter directly processes the code, making development faster and easier.

When a Python program is executed, the interpreter first checks the code for errors and then translates it into an intermediate form called **bytecode**. This bytecode is then run by the **Python Virtual Machine (PVM)**. The interpreter also provides an **interactive mode (REPL – Read, Evaluate, Print, Loop)**, where users can type commands and get immediate results, which helps in learning, debugging, and testing.

Different implementations of the Python interpreter include **CPython** (the default and most widely used), **PyPy** (faster with JIT), **Jython** (runs on Java platform), and **IronPython** (for .NET). Overall, the Python interpreter plays a crucial role in executing code efficiently and providing an easy environment for programmers.

1.5 Data types, Variables, Comments, Operators, expressions; input, processing and output statements:

1. Data Types:

Data types define the kind of values a variable can store. In Python, data types are dynamic, meaning the type is assigned automatically at runtime. The major built-in data types include:

1. **Numeric Types:**
   Integers (`int`), floating-point numbers (`float`), and complex numbers (`complex`).

2. **String Type:**
   `str` represents a sequence of characters enclosed in quotes.

3. **Boolean Type:**
   `bool` stores either `True` or `False`, often used in decision-making.

4. **Sequence Types:**
   List: Ordered and mutable (e.g., `[1,2,3]`).
   Tuple: Ordered and immutable (e.g., `(1,2,3)`).
   Range: Represents a sequence of numbers.

5. **Mapping and Set Types:**
   Dictionary (`dict`): Stores key–value pairs.
   Set: Unordered collection of unique elements.

Python's rich variety of data types makes it suitable for handling simple to complex data structures.

2. Variables:

A **variable** is a name given to a memory location that stores data. In Python, variables are created automatically when a value is assigned using the `=` operator.

Characteristics:

1. **Dynamic Typing:**
   Variable type changes based on the assigned value.

2. **No Declaration Required:**
   Python does not require explicit type declaration.

3. **Naming Rules:**
   Must start with a letter or underscore; cannot start with a digit; cannot use keywords.

4. **Multiple Assignments:**
   Python allows assigning multiple variables in one line (e.g., `a, b = 10, 20`).

5. **Variable Reassignment:**
   Variables can be reassigned to different data types anytime.

Variables help store, update, and manipulate data during program execution.

3. Comments:

Comments are non-executable lines used to explain code. Python ignores comments during execution.

Types:

1. **Single-line Comments:**
   Start with `#` and are used to describe a single line or statement.

2. **Multi-line Comments:**
   Written using triple quotes (`''' ... '''` or `""" ... """`), usually for large explanations or documentation.

Uses:

Improve code readability
Describe logic, purpose, or functionality
Help debugging by disabling lines temporarily

Comments make programs easier to understand and maintain.

4. Operators:

Operators are symbols used to perform operations on variables and values.

Types of operators in Python:

1. **Arithmetic Operators:** `+`, `-`, `*`, `/`, `%`, `//`, `**`
2. **Relational/Comparison Operators:** `>`, `<`, `>=`, `<=`, `==`, `!=`
3. **Logical Operators:** `and`, `or`, `not`
4. **Assignment Operators:** `=`, `+=`, `-=`, `*=`, `/=`, etc.
5. **Bitwise Operators:** `&`, `|`, `^`, `~`, `<<`, `>>`
6. **Membership Operators:** `in`, `not in`
7. **Identity Operators:** `is`, `is not`

Operators allow Python to perform calculations, comparisons, logic decisions, and more.

5. Expressions:

An **expression** is a combination of values, variables, operators, and function calls that produce a result.

Examples:

Arithmetic: `a + b * 3`
Boolean: `x > 10 and y < 5`
String Expressions: `"Hello" + "World"`

Properties:

Expressions are evaluated by Python to generate a value.
Used in assignments, conditions, loops, and function arguments.
Form the building blocks of any program.

Expressions help perform computations and determine program behavior.

6. Input, Processing, and Output Statements:

Input:

Used to take data from the user using the **`input()`** function.
Example: `name = input("Enter your name: ")`

Processing:

This involves performing operations or computations using variables, expressions, and logic.

Example:

```python
total = a + b
```

Output:

Displayed to the user using `**print()**` function.

Example: `print("Total =", total)`

These three steps form the basic structure of most programs:

1. **Input:** Get data
2. **Processing:** Perform calculations or logic
3. **Output:** Display result

1.6 Control Structures: loops and decision:

Control structures in Python allow the program to make decisions and repeat actions based on conditions. They determine the flow of execution.

1. Decision Control (Selection Statements):

Decision-making statements allow the program to choose different paths based on a condition. Python provides:

`**if**` **statement:** Executes a block when the condition is true.

`**if–else**` **statement:** Executes one block if the condition is true, otherwise another block.

`**elif**` **ladder:** Used when multiple conditions need to be checked in sequence.

These help in performing conditional operations, validations, and branching logic.

Example:

```python
if marks >= 40:
    print("Pass")
else:
    print("Fail")
```

2. Loop Control (Iteration Statements):

Loops are used to execute a block of code repeatedly.

`**for**` **loop:** Iterates over a sequence such as list, tuple, string, or range.

Example:

```python
for i in range(5):
print(i)
```

`**while**` **loop:** Repeats a block as long as a condition remains true.

Example:

```python
while n > 0:
n -= 1
```

Loops help in tasks like traversing data, repeated calculations, and automation.

Conclusion:

Decision structures control which statements execute, while loop structures control how many times statements repeat. Together, they form the fundamental logic-building blocks of Python programs.